Software Requirements Specification

Team V1 - AI & Motion Recognition

Requirement Analysis - Updated Version
Project: Limb Motion Recognition and Assistant
Prepared for: Distributed Software Development 2025-2026
Prepared by: Team V1 - AI & Motion Recognition
Original date: 3 April 2026
Updated date: 6 May 2026
Version: 1.4

> This version keeps V2 as the only direct communication point for Team V1. It also clarifies the practical Generation 1 scope after the latest team-leader alignment: for now, the patient uploads two angle values inside the current jointAngles format, and V1 produces a plain-text recommendation through V2. A future generation may use the richer targetAngles / sensorData / errors format for a more complete IMU-based AI pipeline.

Revision History

Date	Author	Description
3 Apr 2026	Team V1	Initial Requirement Analysis. V1 expected direct input from S2 and output to V2.
6 May 2026	Team V1	Version 1.2 aligned V1 with Total System Design and Total Interface Specification. V1 input was moved to V2 and S2 became an indirect upstream source.
6 May 2026	Team V1	Version 1.3 added a Generation 1 / Future Generation distinction. Generation 1 uses the current deployed V2 API; the future generation keeps the richer targetAngles + sensorData + errors direction.
6 May 2026	Team V1	Version 1.4 clarifies that Generation 1 uses two patient-uploaded angle values inside jointAngles,

Date	Author	Description
		and that the immediate V1 output is a plain-text recommendation sent to V2.

Document Overview

Field	Value
Area	Server
Team	V1 - AI & Motion Recognition
Main role	Read motion/session measurements from V2, analyse the available angle values, and provide recommendation/results back to V2.
Main dependencies	V2 for direct server-side input/output; S2 and M1 as indirect upstream sources of formatted data; M1/M2 as downstream consumers through V2.
Main deliverables	Generation 1 two-angle analysis, plain-text recommendation output to V2, preprocessing and validation logic, future IMU-ready pipeline, processing metadata, and error/status handling when supported.

0. Background

This document describes the updated requirement analysis for Team V1 in the Limb Motion Recognition and Assistant project. The project is developed in a distributed setting between UTAD and Jilin University.

Team V1 is responsible for the AI and motion-recognition part of the Server pipeline. V1 is not responsible for physical sensors, the patient mobile application, the clinical dashboard, or the V2 database implementation.

The current integrated architecture places V2 as the central server-side hub. V1 reads measurements from V2, processes them, and writes analysis results or recommendations back to V2. S2 remains the upstream source of formatted sensor information, but V1 does not communicate directly with S2.

For the current Generation 1 implementation, the latest team-leader alignment states that the patient uploads two angle values. These values are handled through the current V2 jointAngles format. V1 uses these two angles to create a plain-text recommendation. The doctor may also provide plain-text recommendation information. V1 output should therefore be treated as AI-assisted support information, not as a final clinical decision.

0.1 Purpose of this document

- Define the updated requirement analysis of Team V1 using the shared SRS style.
- Make clear what V1 expects to receive from V2.
- Make clear what V1 provides back to V2.
- Clarify that S2 is an indirect upstream dependency through M1 and V2.
- Separate the current Generation 1 implementation from the future richer IMU-based extension.
- Clarify that Generation 1 uses two patient-uploaded angle values and plain-text recommendations.
- Define the main V1 use cases, interfaces, and output delivery approach.

0.2 Scope of Team V1

- Read session measurements from V2 using the agreed V2 interface.
- For Generation 1, validate and preprocess two patient-uploaded angle values contained in jointAngles, plus isCorrect, timestamp, and sessionId from the current V2 API.
- For Generation 1, generate a simple plain-text recommendation using angle-based or rule-based logic.
- For a future generation, prepare for targetAngles, sensorData, timestamps, and errors when the richer interface is implemented.
- Run V1 analysis, inference, comparison, or rule-based recommendation logic according to the available data.
- Create structured recommendation/results and send them to V2 through POST /recommendations.
- Keep useful processing metadata, warnings, confidence values, and error states when supported by the interface.

0.3 Team boundaries

- S1 captures raw IMU data from sensors.
- S2 validates and formats sensor data and sends formatted data to M1. If target-angle values are produced by S2, V1 accesses them only after they are stored or exposed by V2.
- M1 acts as the App-side gateway to the Server and uploads measurements to V2.
- V2 stores measurements and exposes server-side API endpoints.
- V1 reads measurements from V2 and writes analysis/recommendation results back to V2.
- M1 and M2 consume V1 results indirectly through V2.
- Doctors may provide recommendation text through the appropriate Monitor/Server workflow, but V1 does not communicate directly with the doctor interface.

0.4 Implementation note - Generation 1 and future generation

For Generation 1, Team V1 will use the current deployed V2 API format. In this version, the patient uploads two angle values, and V2 exposes them inside the jointAngles object. Example:

```json
{
 "sessionId": 1,
 "jointAngles": {
  "knee": 45.0,
  "hip": 30.0
 },
 "isCorrect": false,
 "timestamp": "2026-05-05T10:00:00.000Z"
}
```

In this first generation, V1 does not receive raw IMU data and does not run a complete AI motion-recognition model. V1 uses the two uploaded angle values as input and produces a simple plain-text recommendation using rule-based or angle-based logic. This keeps the first integration stable and avoids requesting endpoint changes from V2 before the first presentation.

For a future generation, Team V1 expects the interface to evolve toward a richer format aligned with the integrated system documentation, using targetAngles, sensorData, and errors. This future format will be needed for a more complete AI-based motion recognition pipeline using raw IMU-style data.

Therefore, this requirement analysis supports two levels:

1. **Generation 1:** integration with the current V2 API using two angle values inside jointAngles and plain-text recommendation output;
2. **Future generation:** extended IMU-based pipeline using full sensor data, target/reference angle information, errors, sensor mapping, and labels when available.

In both levels, V1 communicates directly only with V2.

0.5 Cross-team data exchange

0.5.1 What V1 expects to receive

Source	Item	Why V1 Needs It	Notes
V2 - Backend API & Storage	Session measurements through GET /measurements/:sessionId	Main input path for V1 analysis.	Official V1 input path in the integrated system design.
V2 - Backend API & Storage	Generation 1 jointAngles / joint_angles	Current input X for V1. Contains two patient-uploaded angle values, for example knee and hip.	Exact angle names should be confirmed by V2/M1 if they differ from the example.
V2 - Backend API &	Generation 1	Current API field.	V1 should not use it

Source	Item	Why V1 Needs It	Notes
Storage	isCorrect / is_correct	May represent an existing classification/status field.	as the answer for its own decision. It can be treated as state/label depending on the implementation.
V2 - Backend API & Storage	Generation 1 timestamp and sessionId / session_id	Needed to identify and trace the session and order measurements.	Required for linking V1 output back to the correct session.
V2 - Backend API & Storage	Future fields: targetAngles / target_angles	Future angle/reference information for V1.	Exact semantics still need confirmation.
V2 - Backend API & Storage	Future fields: sensorData / sensor_data	Input for future IMU-based motion recognition.	May include sensorId, accX/Y/Z, gyroX/Y/Z, roll, pitch, yaw, and timestamp. Depends on future V2/S2 alignment.
V2 - Backend API & Storage	Future fields: errors	Data-quality and sensor-error information.	Used to filter invalid data, create warnings, or reduce confidence.
S2 - Data Acquisition, indirectly through M1/V2	Field meaning, sensor placement, target-angle meaning, sampling rules, and error rules	Needed to interpret future sensor data correctly.	S2 is not a direct V1 interface, but its definitions are still necessary.
M1/M2 through project alignment	Required recommendation display needs	Needed so V1 produces useful output for patient and clinician views.	V1 still sends results to V2, not directly to M1/M2.

0.5.2 What V1 will provide

Receiver	Item	Purpose	Notes
V2 - Backend API & Storage	Plain-text recommendation / analysis result	Allows V2 to store and expose V1 output to M1 and M2.	Official output path: POST /recommendations.
V2 - Backend API & Storage	Movement name and confidence	Allows downstream systems to understand what was	Required by current recommendation schema if supported.

Receiver	Item	Purpose	Notes
		analysed and how confident V1 is.	
V2 - Backend API & Storage	Suggestion status	Allows M2 or other workflows to manage recommendation review state.	Current allowed values may include pending, accepted, rejected.
V2 - Backend API & Storage	Warnings, metadata, and detailed text when supported	Improves traceability and user-facing explanations.	Some fields may require future V2 schema extension.
M1/M2, indirectly via V2	AI-assisted feedback/recommendation information	Allows patient and clinician interfaces to display V1 results.	No direct V1-to-M1 or V1-to-M2 interface.
Doctor workflow, indirectly through V2/M2	Doctor plain-text recommendation or review	Allows medical recommendation text to coexist with V1 AI-assisted text.	V1 output is supportive and should not replace doctor judgement.

0.5.3 Interface definition summary

Interface	Direction	Owners	Main Content
IF-V2-V1	V2 -> V1	V2 and V1	Generation 1 measurements: two uploaded angle values inside jointAngles / joint_angles, isCorrect / is_correct, timestamp, sessionId / session_id. Future measurements: targetAngles / target_angles, sensorData / sensor_data, errors, timestamps, session identifiers, and related reference data.
IF-V1-V2	V1 -> V2	V1 and V2	Recommendation or analysis result: sessionId,

Interface	Direction	Owners	Main Content
			movement, confidence, status, and plain-text recommendation when supported. Future metadata may be added later.
Upstream data definitions	S2/M1 -> V2 -> V1	S2, M1, V2, V1	S2 defines sensor data meaning, target-angle computation, sensor mapping, and error representation; V1 receives any needed definitions through project alignment and V2 exposure.

0.5.4 Output delivery approach

- V1 will send its output to V2 using the agreed V2 endpoint, currently POST /recommendations.
- For Generation 1, the main output should be a plain-text recommendation based on the two uploaded angle values.
- Each result should include a session identifier, movement name, confidence value, and status when supported by the V2 schema.
- When supported, V1 should also include warnings, model/rule version, processing timestamp, and human-readable recommendation text.
- The output should be easy for V2 to store and easy for M1/M2 to consume through V2.
- V1 should not write directly to M1, M2, S1, or S2.

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1. Use Cases

Login, registration, account management, sensor connection, doctor review, and final user-interface actions are not included here because they belong more naturally to V2, S1/S2, M1, or M2. The use case diagram should show V2 as the main external actor for V1.

ID	Use Case	Primary Actor	Priority
UC-V1-01	Process Session Measurements from V2	V2 - Backend API & Storage	High
UC-V1-02	Analyse Uploaded Angle Values and	None (included internal V1 use case)	High

ID	Use Case	Primary Actor	Priority
	Future Sensor Data		
UC-V1-03	Provide Plain-Text Recommendation Results to V2	V2 - Backend API & Storage	High

2.1 Process Session Measurements from V2

2.1.1 Basic Info

Field	Value
Reference to Use Case	UC-V1-01
Version	1.4
Created	3 Apr 2026; updated 6 May 2026
Authors	Team V1 - AI & Motion Recognition
Source	Server area / Team V1
Actors	Primary actor: V2 - Backend API & Storage. Indirect upstream source: S2 through M1 and V2.
Goal	Receive session measurements from V2 and prepare them for V1 analysis.
Summary	This use case starts when V1 requests or receives available measurements for a session from V2. Generation 1 uses the current V2 API format, where the patient uploads two angle values inside jointAngles. A future generation may use the richer targetAngles + sensorData + errors format.
Trigger	A session is ready to be analysed, or V2 provides measurements for a session.
Frequency	Whenever V1 needs to analyse a rehabilitation session.
Precondition	V2 has a session identifier and available measurements; the V2 API is reachable; V1 processing service is available.
Postconditions	Cleaned and normalized data is available for V1 analysis, or a structured validation error is produced.

Flow Snapshot: Request Measurements -> Validate Payload -> Normalize Fields -> Prepare Input -> Pass to Analysis

2.1.2 Basic Flow

Actor	System (V1 Module)
V1 calls V2 using GET /measurements/:sessionId.	Receive measurement data for the requested session.
V2 returns available Generation 1 fields: joint_angles, is_correct, timestamp, and session_id.	Check whether the payload follows the current V2 response structure.
In Generation 1, joint_angles contains two uploaded angle values, for example knee and hip.	Validate the two angle values, timestamp, and session identifier.
In a future generation, V2 may return target_angles, sensor_data, errors, timestamp, and session_id.	Normalize the available fields for internal processing.
	Handle optional or missing future fields without breaking Generation 1 processing.
	Prepare cleaned data for analysis/recommendation logic.

2.1.3 Alternative Flow

Situation	System (V1 Module)
V2 API is unavailable.	Return an integration error and stop processing.
Measurements are empty.	Return no-data status.
Generation 1 joint_angles are missing or invalid.	Return validation warning or partial-processing status.
Only one of the two expected angle values is available.	Process partially if allowed, otherwise return an insufficient-data status.
Future target_angles are missing or unclear.	Use Generation 1 fields if available, or mark the result as assumption-dependent.
Future sensor_data is empty.	Continue with angle-based analysis if angle data is available.
errors are present.	Use them as data-quality warnings and adjust confidence or processing status.

2.2 Analyse Uploaded Angle Values and Future Sensor Data

2.2.1 Basic Info

Field	Value
Reference to Use Case	UC-V1-02
Version	1.4

Field	Value
Created	3 Apr 2026; updated 6 May 2026
Authors	Team V1 - AI & Motion Recognition
Source	Server area / Team V1
Actors	None as a direct external actor. This is an internal/included use case invoked by UC-V1-01.
Goal	Analyse the two uploaded angle values in Generation 1 and prepare for future sensor-data-based analysis.
Summary	Generation 1 uses two angle values inside joint_angles for rule-based or simple angle-based recommendation logic. A future generation can use target_angles and sensor_data for richer IMU-based motion recognition if complete data and mapping are available.
Trigger	Cleaned measurements are available after preprocessing.
Frequency	Whenever a session is processed by V1.
Precondition	Valid Generation 1 angle data exists, or future target/sensor data exists. Required rules/configuration are available.
Postconditions	A recommendation candidate, comparison result, raw prediction, or structured warning is produced.

Flow Snapshot: Read Clean Data -> Apply Rules/Model -> Compare/Interpret -> Build Recommendation -> Send to Post-processing

2.2.2 Basic Flow

Actor	System (V1 Module)
Internal V1 process starts after preprocessing.	Read the available measurement data and metadata.
	For Generation 1, use the two angle values inside joint_angles for angle-based analysis and plain-text recommendation generation.
	Use is_correct only as a current API field; the main V1 output path remains recommendations.
	Classify the angle values using simple rules, thresholds, or rehabilitation guideline ranges

Actor	System (V1 Module)
	when agreed.
	For a future generation, use target_angles, sensor_data, and errors when available and semantically clear.
	Create recommendation output with confidence, status, warnings, and plain text when supported.

2.2.3 Alternative Flow

Situation	System (V1 Module)
joint_angles are available but limited.	Run the Generation 1 angle-based fallback.
Exact names of the two angle fields are unclear.	Use configurable field mapping and request confirmation from V2/M1.
target_angles meaning is unclear.	Mark the result as assumption-dependent and request alignment with S2/V2.
sensor_data exists but sensor-to-joint mapping is missing.	Do not run full IMU-based model; return warning or use angle-based fallback.
errors indicate invalid sensor data.	Exclude affected data or reduce confidence.
Model/configuration is unavailable.	Use rule-based fallback if allowed, or return processing-failure status.

2.3 Provide Plain-Text Recommendation Results to V2

2.3.1 Basic Info

Field	Value
Reference to Use Case	UC-V1-03
Version	1.4
Created	3 Apr 2026; updated 6 May 2026
Authors	Team V1 - AI & Motion Recognition
Source	Server area / Team V1
Actors	Primary actor: V2 - Backend API & Storage. Downstream consumers: M1 and M2 through V2.
Goal	Provide V1 output to V2 in a stable and easy-to-use format.
Summary	This use case covers the handoff from V1 to V2. For Generation 1, V1 sends a plain-text recommendation or analysis result through POST /recommendations so V2 can store

Field	Value
	and expose it to other teams.
Trigger	A V1 recommendation candidate or analysis result is ready for delivery.
Frequency	Whenever V1 finishes a processing routine for a session.
Precondition	The V1-V2 recommendation contract exists and V1 has a result object.
Postconditions	V2 receives a recommendation/result payload or a structured error state.

Flow Snapshot: Package Result -> Check Contract -> POST to V2 -> Store in V2 -> Available to M1/M2

2.3.2 Basic Flow

Actor	System (V1 Module)
A V1 analysis routine finishes.	Package sessionId, movement, confidence, status, and plain-text recommendation when supported.
	Add optional warning/metadata fields only if supported by V2.
	Validate the payload against the agreed V2 recommendation schema.
V1 sends POST /recommendations to V2.	V2 stores the recommendation/result.
M1 or M2 later requests recommendations from V2.	V2 exposes V1 results through its own API.

2.3.3 Alternative Flow

Situation	System (V1 Module)
V2 recommendation endpoint is unavailable.	Return integration error and retry later if supported.
Required output fields are missing.	Reject the internal payload before sending to V2.
V2 does not support a dedicated recommendation text field.	Send the closest supported text/status field and keep detailed text internally or pending.
V2 does not support extended fields.	Send only the required fields and keep detailed metadata internally or pending.
Analysis confidence is low.	Send low-confidence status or warning if supported.
Doctor recommendation differs from V1 recommendation.	Treat both as separate recommendation sources if the platform supports it; doctor decision should have priority in clinical

Situation	System (V1 Module)
	interpretation.

3. Interface and Data Exchange Summary

3.1 IF-V2-V1 - Measurements from V2 to V1

Field / Topic	Generation	What V1 Expects
session_id / sessionId	Both	Session identifier used to trace the measurements and later recommendation output.
timestamp	Both	ISO 8601 timestamp for measurement ordering and time alignment.
joint_angles / jointAngles	Generation 1	Current deployed angle object used by the first V1 implementation. It contains two patient-uploaded angle values, for example knee and hip.
is_correct / isCorrect	Generation 1	Current deployed boolean field. V1 should not treat this as the main long-term output path unless V2 confirms it.
target_angles / targetAngles	Future generation	Future angle stream. Each item should include timestamp, angle_id/angleID, and angle when implemented.
sensor_data / sensorData	Future generation	IMU-like data for future AI processing. Expected fields include timestamp, sensorId, accX/Y/Z, gyroX/Y/Z, roll, pitch, and yaw.
errors	Future generation	Error events or warnings that describe invalid, missing, or unreliable input.
reference/session context	Future generation	When available, V1 needs sensor-to-joint mapping, exercise type/payload status, and reference-rule information.

3.2 IF-V1-V2 - Recommendations from V1 to V2

Output Block	What V1 Should Send to V2
sessionId	The session being analysed.
movement	Movement or exercise analysed by V1, for example knee flexion.
confidence	Confidence score between 0.0 and 1.0, when supported.
status	Initial recommendation status, normally pending unless otherwise agreed.
plain-text recommendation	Main Generation 1 recommendation output. Field name should follow V2's agreed schema, for example recommendationText, notes, or equivalent if supported.
Future optional details	Joint name, angle, phase, warnings, model/rule version, or processing timestamp if V2 supports them.

3.3 Generation 1 and future V1 input/output view

Version	Input X	Output Y	Notes
Generation 1 - current V2 API	jointAngles / joint_angles containing two patient-uploaded angle values, plus timestamp, sessionId / session_id, and current isCorrect / is_correct field	plain-text recommendation/result with movement, confidence, and status through POST /recommendations when supported	Supports first integration and rule-based analysis without changing V2 endpoints.
Future generation - richer interface	targetAngles / target_angles, sensorData / sensor_data, errors, timestamps, and sensor mapping	recommendation/ result, possible detailed predictions, warnings, and metadata	Requires complete sensor data, sensor-to-joint mapping, labels, and V2 schema support.

3.4 Delivery rules

- V1 input should come through V2, using the V2 measurements API.
- V1 output should go to V2, using the V2 recommendations API.
- For Generation 1, V1 should use the current V2 fields and avoid requesting endpoint changes before the first presentation.
- For Generation 1, V1 should treat the two uploaded angle values as the current input X.

- For Generation 1, V1 should treat the plain-text recommendation as the current output Y.
- For a future generation, V1 should be ready to support targetAngles, sensorData, and errors when V2 and S2 align the schema.
- Field names should follow V2 conventions: camelCase in requests and snake_case in responses.
- S2 remains important for the meaning of sensor and future target-angle fields, but V1 should not implement a direct S2 interface.

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4. Functional Requirements Summary

ID	Requirement	Notes
FR1	The V1 module should receive session measurements from V2.	Input comes through GET /measurements/:sessionId.
FR2	The V1 module should support the current Generation 1 V2 measurement format.	Current fields: jointAngles / joint_angles, isCorrect / is_correct, timestamp, sessionId / session_id.
FR3	The V1 module should process two patient-uploaded angle values in Generation 1.	These values are expected inside jointAngles; example fields may be knee and hip.
FR4	The V1 module should validate required fields before processing starts.	Examples: session_id / sessionId, timestamp, and available angle fields.
FR5	The V1 module should transform valid input into analysis-ready data.	Preprocessing, normalization, timestamp alignment, and feature preparation.
FR6	The V1 module should support a Generation 1 angle-based analysis path using jointAngles.	Useful for early integration and rule-based recommendations.
FR7	The V1 module should generate a plain-text recommendation for Generation 1.	Output should be sent through POST /recommendations using the fields supported by V2.
FR8	The V1 module should prepare for a future richer interface using targetAngles, sensorData, and errors.	Requires V2/S2 alignment and later schema support.
FR9	The V1 module should	Requires mapping, sampling

ID	Requirement	Notes
	support future IMU-based analysis using sensorData when available.	rules, labels, and complete sensor data.
FR10	The V1 module should include confidence and status values with each result when supported.	Required or useful for downstream display and review.
FR11	The V1 module should report processing errors, missing data, or low-confidence results consistently.	Important for traceability and downstream UI behaviour.
FR12	The V1 module should keep useful model/rule and processing metadata when supported.	Examples: rule version, model version, processing timestamp, warnings.

5. Non-Functional Requirements Summary

ID	Requirement	Notes
NFR1	Interface clarity	Input and output formats should be documented so V2, M1, M2, and S2 can understand them quickly.
NFR2	Maintainability	The pipeline should separate V2 client, preprocessing, analysis, recommendation generation, and result publishing.
NFR3	Testability	The module should support repeatable tests with sample measurements from V2.
NFR4	Traceability	Processing status, assumptions, warnings, and technical limitations should be easy to review later.
NFR5	Robustness	V1 should handle missing optional data, unavailable future fields, and error events without crashing.
NFR6	Team readability	Documentation should use simple English and avoid unnecessary complexity.

ID	Requirement	Notes
NFR7	Clinical caution	V1 recommendations should be written as AI-assisted support information, not as final medical decisions.

6. Assumptions and Alignment Notes

- V2 is the official source and sink for V1 server-side communication.
- For Generation 1, the deployed V2 API uses jointAngles + isCorrect + timestamp + sessionId. V1 should support this format for the first integrated version.
- For Generation 1, the patient uploads two angle values, and V1 receives them inside jointAngles through V2.
- The exact names and semantics of the two angle values should be confirmed if they differ from the example knee and hip.
- For Generation 1, V1 produces a plain-text recommendation based on the uploaded angles.
- The doctor may also provide plain-text recommendation information. V1 output is AI-assisted support and should not replace doctor judgement.
- The richer integrated target format is targetAngles + sensorData + errors, with V2 responses expected to use snake_case when implemented.
- The future targetAngles / sensorData / errors format should be treated as a planned extension for later generations, not as a dependency for the first presentation.
- target_angles semantics still need confirmation with S2/V2 before they are used as final model input or comparison reference.
- sensor_data enables a future IMU-based pipeline, but full AI implementation depends on sensor placement, mapping, labels, and complete data availability.
- POST /recommendations is the confirmed V1 output path for the current generation.
- Detailed recommendation text, model metadata, and extra warning fields may require V2 schema support or field-name confirmation.

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7. Open Questions

Question	Why it matters
What are the exact names of the two angle values uploaded by the patient?	V1 needs stable field names or a mapping rule for Generation 1.
Are the two uploaded angles measured values, target/reference values, or patient-entered values?	V1 needs this to know how to interpret them.
Which field should V1 use for the plain-text recommendation in POST /recommendations?	V1 needs to match the V2 API exactly.

Question	Why it matters
Should V1 and doctor recommendations be stored as separate recommendation sources?	This avoids confusing AI-assisted text with clinical text.
Should V1 update isCorrect, or should it only send recommendations for Generation 1?	This decides whether isCorrect remains only an input/status field or also a V1-updated result.
When will V2 move from the Generation 1 schema to the richer targetAngles + sensorData + errors schema?	This decides when V1 can start implementing the full IMU-based pipeline.
What exactly do targetAngles represent: measured angle, target/reference angle, or S2-computed movement angle?	V1 needs this to know whether it is an input feature, comparison target, or already processed result.
Will V2 store and return sensorData in a future generation?	This decides whether the future IMU-based pipeline can be implemented.
Where will sensorJointMapping and payloadStatus be exposed to V1?	V1 needs sensor-to-joint mapping and exercise type to interpret sensor data.
What recommendation fields do M1 and M2 need to display?	V1 output should be useful for patient and clinician interfaces.

8. Summary

This updated requirement analysis reflects the integrated architecture of the project while also respecting the current state of the V2 API. Team V1 treats V2 as the official communication point for both input and output. V1 reads measurements from V2 and sends recommendation/results back to V2.

For Generation 1, V1 will use the current deployed V2 API with jointAngles, isCorrect, timestamp, and sessionId. The latest alignment clarifies that the patient uploads two angle values, and these values are available inside jointAngles. V1 uses these two angle values as input and generates a plain-text recommendation through POST /recommendations when supported by V2.

For a future generation, V1 will prepare for the richer targetAngles + sensorData + errors format. That future version can support a more complete IMU-based AI pipeline, but it depends on complete data, sensor mapping, labels, agreed semantics, and V2 schema support.

This update keeps the removal of the previous direct S2 -> V1 assumption. S2 remains an important upstream source of data definitions, but V1 should not implement direct communication with S2 in the integrated system.